

Learning Lean 4

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Abstract

Working notes and exercises from learning the Lean 4 theorem prover, following *An Introduction to Lean 4* by Cosme Llópez and Gong. Covers basic syntax and natural-deduction proofs, paired with a runnable Lean library.

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1 Basic Syntax

1.1 Basic syntax

The runnable version of this chapter lives in `LearningLean/BasicSyntax.lean`.

1.1.1 Comments

```
-- This is a single-line comment.
```

```
/-  
This is a multi-line comment.  
Here is another line.  
-/
```

1.1.2 Inspecting terms and types with `#check`

`#check` asks Lean for the type of an expression without evaluating it.

```
#check true          -- Bool  
#check 42            -- Nat  
#check "Hello, World!" -- String  
#check Type          -- Type 1  
#check Nat           -- Type  
#check ['h', 'e', 'l', 'l', 'o'] -- List Char
```

1.1.3 Printing definitions with `#print`

`#print` shows how something is defined.

```
#print Bool
```

Type itself cannot be printed with `#print`, because it is a type rather than a definition.

1.1.4 Definitions and functions

```
def pi : Float := 3.1415926
```

```
#check pi
```

```
-- An anonymous function (lambda) taking two natural numbers.
```

```
#check λ (a b : Nat) => a + b
```

2 Natural Deduction

2.1 Natural deduction

The Lean proofs for this chapter live in `LearningLean/NaturalDeduction.lean`, inside the `Exercises` namespace.

2.1.1 5.5 Reduction to the absurd. $P \Rightarrow Q, \neg Q \vdash \neg P$

This is a very useful technique. Validity of $P \Rightarrow Q, \neg Q \vdash \neg P$ is proved with:

1	$P \Rightarrow Q$	
2	$\neg Q$	
3	$\begin{array}{ l} P \end{array}$	H
4	$\begin{array}{ l} Q \end{array}$	$\Rightarrow$ 1,3
5	$\begin{array}{ l} \neg Q \end{array}$	IT 2
6	$\neg P$	I $\neg$ 3,4,5

The sub-derivation (lines 3–5) discharges the hypothesis P : assuming P , [$\Rightarrow$ -elimination] on lines 1 and 3 gives Q , while line 2 reiterates $\neg Q$. The contradiction lets us introduce $\neg P$ ($\neg$ -introduction over lines 3, 4, 5).

In Lean this is theorem `T55`, where $\neg P$ is definitionally $P \rightarrow \text{False}$, so the whole proof is just function composition:

```
theorem T55 (h1 : P → Q) (h2 : ¬Q) : ¬P :=  
  (h2 ∘ h1)
```

2.1.2 Conjunction introduction. $P, P \Rightarrow Q \vdash P \wedge Q$

Theorem `T51` builds the pair (proof of P , proof of Q) directly, using the anonymous-constructor notation `{_, _}`:

```
theorem T51 (h1 : P) (h2 : P → Q) : P ∧ Q :=  
  {h1, h2 h1}
```